Definitions

An example from linear algebra

Definition: A set $\{v_1 \ldots, v_n\}$ of elements of a vector space V is called *linearly independent* if, for any set $a_1, \ldots, a_n$ of scalars, if $\sum_{i=1}^{n} a_i v_i = 0$ then $a_i = 0$ for all $i = 1, \ldots, n$. A set of vectors that is not linearly independent is called *linearly dependent.*

The only solution to $\sum a_i v_i = 0$ is if all $a_i = 0$

If $a_1, \ldots, a_n$ are a set scalars, not all zero, then $\sum a_i v_i \neq 0$.

$\forall (a_1, \ldots, a_n)$, $\exists i$ s.t. $a_i \neq 0$, $\sum a_i v_i \neq 0$.

Theorem: The set of vectors $\{(1,3),(2,2)\}$ is linearly independent in $\mathbb{R}^2$.

Does there exist (a_1, a_2) NOT BOTH zero such that

$$a_1(1,3) + a_2(2,2) = 0$$

$$\begin{cases} a_1 + 2a_2 = 0 \\ 3a_1 + 2a_2 = 0 \end{cases}$$

$$\Rightarrow \quad a_1 = a_2 = 0$$

Theorem: The set of vectors $\{(1,1,1),(2,2,2),(1,3,2)\}$ is linearly dependent in $\mathbb{R}^3$.